



Research paper

A Local-Global Fusion Vision Mamba UNet Framework for medical image segmentation

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ABSTRACT

As a State Space Model (SSM) that achieves long-range dependency modeling with linear computational complexity, Mamba demonstrates significant efficiency advantages in medical image segmentation. However, while Mamba-based methods enable long-range modeling with linear complexity, their global dependency mechanisms often lead to local feature attenuation, particularly affecting the processing of complex anatomical structures. Existing multi-scale fusion methods also exhibit limited compatibility with State Space Models. To address these challenges, this paper proposes the Local-Global Fusion Vision Mamba UNet (LGFVM-UNet) framework. Its core innovation lies in the Dynamic Gating-enhanced Local-Global Fusion Visual State Space (LGF-VSS) block, which enables the synergistic modeling of global context and local details. Additionally, we designed a Multi-level Cross-scale Feature Fusion Block (MCFB) that enhances multi-scale feature representation through bidirectional resampling and spatial-channel dual attention mechanisms. Additionally, we propose a Gradient Statistics-based Adaptive Hierarchical Loss that dynamically adjusts multi-level supervision weights to optimize the learning process. The proposed method is experimentally validated on five public medical image segmentation datasets spanning diverse imaging modalities and anatomical structures. Results demonstrate that our approach outperforms state-of-the-art methods, excelling in long-range dependency modeling, local detail capture, and multi-scale feature fusion. The source code of our work is available at <https://github.com/NicoleDyson/LGFVM-UNet>.

1. Introduction

Medical image segmentation is a cornerstone of clinical workflows, directly influencing diagnostic accuracy and treatment planning. Unlike natural scenes, medical images present unique challenges such as low inter-class variance, blurred boundaries, and significant scale variations of anatomical structures. While Convolutional Neural Networks (CNNs), particularly the UNet (Ronneberger et al., 2015) architecture, have achieved success through local feature extraction, they struggle with global context modeling due to limited receptive fields. Vision Transformers (ViTs) address this via self-attention but incur quadratic

computational costs and often fail to preserve high-frequency local details essential for precise boundary delineation. Recently, State Space Models (SSMs), notably Mamba (Gu and Dao, 2023), have emerged as a promising alternative, offering global receptive fields with linear computational complexity.

Despite its efficiency, directly applying Mamba to medical imaging reveals two critical limitations, which motivate the design of our framework. First, the “Global Dominance” inherent in Mamba often leads to local feature attenuation. The process of flattening 2D images into 1D sequences and processing them via recurrent transitions can

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dilute spatial relationships between immediate neighbors. In clinical scenarios, such as delineating tumor margins, local texture and edge information are paramount. Second, a significant semantic gap exists between Mamba's linear scanning and the decoder's spatial reconstruction. Standard skip connections fail to align these representations, resulting in suboptimal information flow. To resolve these conflicts, we propose LGFVM-UNet, where the technological integration is driven by functional necessity to address Mamba's architectural biases:

- **Local-Global Fusion Visual State Space (LGF-VSS) Block:** To counteract local detail loss, we introduce parallel multi-scale convolutional branches. A key innovation is the QuadGate module, a dynamic gating mechanism that assigns pixel-wise weights to different branches. By adaptively prioritizing Mamba in uniform tissue areas for global consistency and convolutional branches at complex boundaries for local precision, QuadGate ensures an optimal balance between global context and local features.
- **Multi-level Cross-scale Feature Fusion Block (MCFB):** Replacing simple skip connections, the MCFB serves as a bridge for semantic alignment. It leverages bidirectional resampling and spatial-channel dual attention to ensure that encoder details are semantically aligned with the decoder's global context. This allows the decoder to integrate the entire encoding history, effectively mitigating the semantic gap between 1D scanning and 2D reconstruction.
- **Gradient Statistics-based Adaptive Hierarchical Loss:** Standard static loss weighting cannot adapt to the shifting focus of learning across different scales. We propose an adaptive strategy that monitors real-time gradient magnitudes to automatically adjust supervision weights. This stabilizes training and ensures the model focuses on the most difficult semantic levels as optimization progresses.

2. Related work

2.1. Evolution of UNet architecture

The UNet architecture revolutionized medical image segmentation by introducing a symmetric encoder-decoder design coupled with skip connections. This design elegantly resolves a core tension in the task: semantic classification necessitates downsampling to capture context, while precise spatial localization requires upsampling to recover detail.

Early enhancements focused on structural density. UNet++ (Zhou et al., 2018) introduced nested and dense skip connections to reduce the semantic gap between the encoder and decoder sub-networks. By aggregating features at various semantic levels, it improved the segmentation of objects of varying sizes. Attention U-Net (Oktay et al., 2018) introduced gating signals in the skip connections, allowing the model to suppress irrelevant regions (background noise) and focus on target structures.

Recent advancements have addressed specific limitations in the standard UNet framework. For ultra-high-resolution images, DDU-Net (Cheng et al., 2022) introduced domain decomposition strategies that divide input images into non-overlapping patches for independent processing while enabling contextual exchange between sub-images, significantly reducing memory requirements without compromising accuracy. For local detail enhancement, CBAM-AG U-Net (Vanaja and Prakasam, 2025) integrated Convolutional Block Attention Module gates into skip connections to suppress background noise and amplify lesion features, substantially improving segmentation accuracy for microstructures. However, these improvements remained bound by the convolution operation. While DC-UNet (Lou et al., 2021) and N-Net (Liang et al., 2023) enhanced feature extraction through dual-channel and dual-encoder designs respectively, they could not escape the limited receptive field paradox: enlarging the kernel size increases

computational cost quadratically, while keeping it small limits context. Multi-path designs like LadderNet (Zhuang, 2018) addressed topological discontinuities but still relied on local operators. The consensus in the field became clear: to achieve the next leap in performance, the model must break free from the local receptive field of the convolution.

2.2. Application of Transformers in medical image segmentation

Transformers have significantly impacted medical image segmentation by overcoming CNN limitations in modeling long-range dependencies. Early approaches like TransUNet (Chen et al., 2021) embedded Transformers into the encoder to model global relationships via serialized image patches. However, their quadratic computational complexity limited applications in high-resolution images. Subsequent hierarchical designs, such as Swin-UNet (Cao et al., 2022), improved efficiency by employing windowed self-attention within local regions and achieving global interaction via window shifting.

Despite their global modeling capabilities, pure Transformer architectures often struggle with fine boundary delineation due to insufficient local feature capture. This limitation has led to the development of hybrid architectures that combine the strengths of both paradigms. SSFormer (Shi et al., 2022) synergizes a Progressive Local Decoder with a pyramid Transformer encoder to gradually aggregate local details and global semantics. TransFuse (Zhang et al., 2021) adopts a parallel dual-branch design where a CNN branch extracts local texture features and a Transformer branch models global context, aligned through a Bidirectional Feature Fusion module.

More efficient approaches have been developed to address computational constraints. MALUNet (Ruan et al., 2022) introduces a lightweight multi-attention UNet that balances performance with minimal parameters. LeViT-UNet (Xu et al., 2023) integrates a LeViT Transformer module into the UNet architecture to trade off accuracy and efficiency, making faster encoders for medical image segmentation through optimal feature extraction. AFTER-UNet (Yan et al., 2022) handles 3D volumetric data through an axial fusion mechanism. This calculates attention separately along the axial axis and within slices before fusing the context.

While these hybrid approaches have shown promising results, they still face fundamental trade-offs between computational complexity and detail preservation. The quadratic complexity of self-attention mechanisms in Transformers remains a bottleneck for high-resolution medical images, creating an opportunity for more efficient sequence models like Mamba that offer linear computational scaling.

2.3. Advances in State Space Models (SSMs)

State Space Models (SSMs), characterized by their linear computational complexity and effective long-range dependency modeling, have recently emerged as a promising paradigm in medical image segmentation. VM-UNet (Ruan et al., 2024) pioneered the integration of Visual State Space (VSS) blocks with UNet's architecture, demonstrating the potential of SSMs for capturing global contextual information efficiently. However, its reliance on fixed scanning strategies limited adaptability to the variable structural orientations in medical images. Building upon VM-UNet, subsequent works have focused on enhancing multi-scale feature fusion and local detail preservation. VM-UNetV2 (Zhang et al., 2024) introduced a Scan-Decoupled Integration (SDI) module to optimize information exchange between different scales. MSVM-UNet (Chen et al., 2024) addressed the directional sensitivity of 2D images by introducing Multi-Scale Visual State Space blocks and a Large Kernel Patch Embedding layer, showing significant improvements on the BraTS2023 dataset.

For specialized learning paradigms, Weak-Mamba-UNet (Wang and Ma, 2024) demonstrated how Visual Mamba can enhance both CNN and ViT performance specifically for scribble-based annotations through a cross-supervisory mechanism employing pseudo labels for

iterative refinement. Recent work (Ma and Wang, 2024) has explored the application of Mamba in semi-supervised learning frameworks through Semi-Mamba-UNet, which combines a Visual Mamba-based architecture with conventional CNN networks. For volumetric data, SegMamba (Xing et al., 2024) extended the application of SSMs to 3D medical image segmentation, outperforming Transformer-based methods by modeling full-volume feature dependencies with hierarchical bidirectional Mamba blocks.

Focusing on efficiency and scanning mechanisms, H-vmunet (Wu et al., 2025) refined the scanning approach by employing a Higher-order 2D Selective Scanning (H-SS2D) to reduce redundant scan paths. LightM-UNet (Liao et al., 2024) addressed computational efficiency by implementing a lightweight Mamba framework with Residual Vision Mamba Layers. UltraLight VM-UNet (Wu et al., 2024) proposed a parallel Vision Mamba approach that significantly reduces model parameters while maintaining performance, achieving remarkable parameter efficiency (0.049M) for skin lesion segmentation. LKM-UNet (Wang et al., 2024) introduced large Mamba kernels with hierarchical and bidirectional designs to establish larger receptive fields. CM-UNet (Liu et al., 2024a) introduced a hybrid CNN-Mamba architecture where a CNN encoder extracts local features and a Mamba-based decoder aggregates global information. Swin-UMamba (Liu et al., 2024b) demonstrated the crucial role of ImageNet-based pretraining for Mamba models in medical segmentation, integrating pretrained VSS blocks with a well-designed decoder for improved performance.

Despite these advances, existing Mamba-based segmentation models still face challenges in balancing global context modeling with local detail preservation. Current approaches typically employ either fixed scanning patterns or simple multi-branch architectures without dynamic adjustment mechanisms, limiting their ability to adapt to the diverse structural patterns in medical images. Additionally, the integration of multi-scale features remains suboptimal, with insufficient attention to bidirectional information flow between encoder and decoder. Our work addresses these limitations through adaptive fusion mechanisms and enhanced cross-scale feature integration. Our LGFVM-UNet differs by integrating explicit local extractors (convolutions) directly into the VSS block, gated dynamically, rather than relying solely on scan path variations.

2.4. Learning-based and anatomy-aware segmentation strategies

Beyond the evolution of backbone architectures, a significant research direction has emerged in medical image segmentation in recent years: the design of specific learning paradigms that explicitly leverage anatomical priors, particularly in semi-supervised and class-imbalanced scenarios.

To address the feature blurring caused by global attention in conventional Transformers, the CASTformer (You et al., 2022) introduced a class-aware adversarial framework. This approach replaces generic query vectors with learnable class embeddings, forcing the model to learn feature prototypes for specific anatomical structures. Simultaneously, it employs an adversarial discriminator to constrain the topological plausibility of segmentation results, effectively mitigating shape distortion in multi-organ segmentation.

In label-scarce semi-supervised settings, ACTION (You et al., 2023b) and its enhanced version ACTION++ (You et al., 2023d) pioneered anatomy-aware contrastive distillation. To tackle the prevalent long-tailed distribution in medical images, ACTION++ proposed an adaptive anatomical contrast mechanism. By dynamically adjusting the temperature coefficient in the contrastive loss, the model can balance gradient contributions according to the learning difficulty of different organs. This shares a profound theoretical connection with the gradient-statistics-based adaptive hierarchical loss proposed in our work: both seek to address imbalance in multi-object learning through dynamic

optimization strategies. Furthermore, MONA (You et al., 2024) demonstrated how global-local contrastive mining can learn anatomical features in an unsupervised manner, further underscoring the importance of cross-scale feature alignment.

Regarding optimization stability and representation refinement, ARCO (You et al., 2023a) addressed gradient variance in segmentation training from a variance reduction perspective, proposing a hierarchical group sampling strategy that significantly improves training stability under extremely limited labels. For boundary representation, MORSE (You et al., 2023c) broke through the limitations of discrete grids by utilizing implicit neural representations and a mixture of stochastic experts to achieve continuous anatomical rendering, thereby refining segmentation boundaries at the sub-pixel level.

In summary, these advanced studies demonstrate that high-performance medical segmentation relies not only on powerful feature extractors but also on a deep understanding of anatomical structures and optimization processes. The design of LGFVM-UNet incorporates the essence of these learning-based methods and internalizes them as architectural priors: we leverage Mamba's linear global modeling capability to replace costly adversarial attention, and employ an adaptive hierarchical loss to substitute complex contrastive pretraining. This ensures anatomical awareness while achieving a significant improvement in computational efficiency.

3. Method

3.1. Overall architecture of LGFVM-UNet

The LGFVM-UNet follows a classic U-shaped topology consisting of a four-stage encoder, a bottleneck, and a four-stage decoder, as illustrated in Fig. 1(a). The framework processes an input image $I \in \mathbb{R}^{H \times W \times 3}$. The encoder processes the input image through four successive scales, with resolutions of $H/4$, $H/8$, $H/16$, and $H/32$. At each scale, features are first extracted by LGF-VSS blocks and then downsampled via Patch Merging. The decoder then restores the resolution using Patch Expanding layers and refines features with LGF-VSS blocks. Crucially, the encoder and decoder are connected not by simple skip connections, but by Multi-level Cross-scale Feature Fusion Blocks (MCFB). The final output is a segmentation mask generated by a 1×1 convolution.

Previous Mamba-based segmentation networks, such as VM-UNet, primarily focus on global context modeling, while others like VM-UNetV2 employ fixed feature fusion strategies. In contrast, our architecture uniquely addresses the dual challenges of balancing global and local features and aligning cross-scale features. The LGF-VSS blocks act as the core computational units, dynamically fusing global context with local details. The MCFB modules complement this by integrating multi-scale features from all encoder stages into the current decoder features, thereby establishing comprehensive cross-scale connections.

3.2. Local-Global Fusion Visual State Space (LGF-VSS) module

The LGF-VSS block serves as the fundamental feature extraction unit. It is designed to capture two complementary types of information simultaneously. The first type is high-frequency details (e.g., texture, edges, and small anatomical anomalies), which are captured via convolutional operations. The second type is low-frequency context (e.g., organ placement, relative geometry, and long-range dependencies), which is modeled via the Mamba branch.

The crucial role of this block is to dynamically arbitrate between these two distinct information sources. For instance, in a region of uniform tissue (e.g., inside the liver), the block should prioritize the Mamba branch to maintain global consistency. Conversely, at a complex boundary like that of a tumor, it should prioritize the Convolutional branch to ensure local precision.

To overcome the local feature attenuation issue in conventional Visual State Space (VSS) modules, we propose the Local-Global Fusion

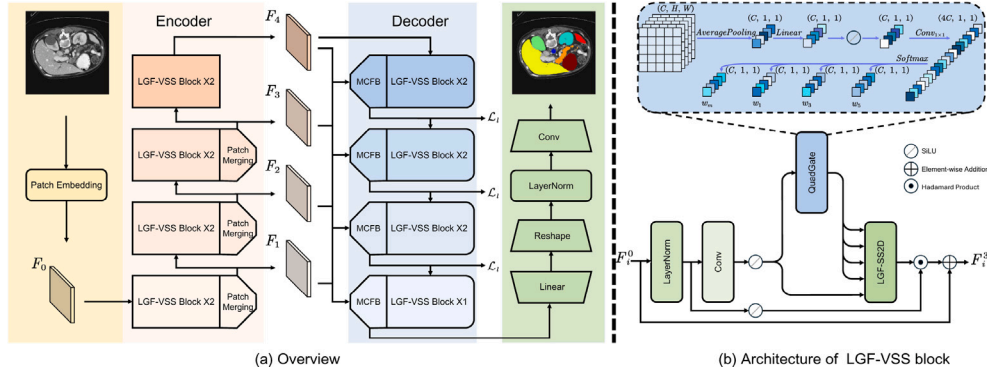


Fig. 1. The overall architecture of our proposed LGFVM-UNet. (a) illustrates the overall architecture of LGFVM-UNet, which consists of a four-stage encoder and a four-stage decoder. (b) presents the structure of the LGF-VSS block. Between the decoder and encoder, we employ the Multi-level Cross-scale Feature Fusion Block (MCFB) to establish multi-scale connections.

Visual State Space (LGF-VSS) module. As illustrated in Fig. 1(b), the LGF-VSS module introduces three key innovations compared to standard VSS blocks: (1) parallel multi-scale convolutional branches for enhanced local feature extraction, (2) a dynamic QuadGate mechanism for adaptive feature fusion, and (3) dual residual connections for optimized gradient flow.

Given an input feature map F_i^0 at stage i , we first apply normalization and activation to prepare it for subsequent operations:

$$F_i^1 = \sigma(\text{Conv}_{3 \times 3}(\text{LN}(F_i^0))), \quad (1)$$

where $\text{Conv}_{3 \times 3}$ represents a standard convolution with 3×3 kernels, $\text{LN}(\cdot)$ denotes layer normalization, and σ is the SiLU activation function. This preprocessing enhances feature representations and stabilizes training.

Unlike fixed-weight fusion strategies in previous works, we introduce the QuadGate module that generates data-dependent weights to adaptively fuse features from different branches. For the processed feature map F_i^1 , the QuadGate module performs:

$$G = \text{Softmax}(\text{Conv}_{1 \times 1}(\sigma(\text{Linear}(\text{AvgPool}(F_i^1))))), \quad (2)$$

where AvgPool reduces spatial resolution to 1×1 while preserving channel dimensions via adaptive average pooling. The Linear projection facilitates cross-channel interaction without altering tensor shape, followed by SiLU activation to enhance nonlinear expressivity. The $\text{Conv}_{1 \times 1}$ expands channel dimensions by $4 \times$ before channel-wise Softmax normalization ensures partition unity.

The generated gating tensor G is split along the channel dimension into four components $\{w_1, w_3, w_5, w_m\}$, corresponding to weights for the 1×1 , 3×3 , and 5×5 convolutional branches, and the Mamba state space branch, respectively. This design allows the network to adaptively emphasize global context or local details based on input characteristics.

The core of our LGF-VSS module is the LGF-SS2D component, which integrates local multi-scale features with global state space representations, as detailed in Fig. 2. Unlike the standard SS2D module, which focuses solely on global context, our design processes F_i^1 through parallel pathways. These pathways include multi-scale convolutional branches (with kernel sizes of 1×1 , 3×3 , and 5×5) for local feature extraction, and a state space branch for global contextual modeling. The fusion process employs the previously generated gating weights for channel-wise adaptive scaling followed by element-wise summation:

$$F_{\text{fused}} = w_m \odot \text{SS2D}(F_i^1) + \sum_{k \in \{1, 3, 5\}} w_k \odot \text{BN}(\text{Conv}_{k \times k}(F_i^1)), \quad (3)$$

where $\odot$ denotes broadcasted element-wise multiplication, and $\text{BN}(\cdot)$ represents batch normalization. The specific details of the SS2D module are illustrated in Fig. 2, where the Scan Expand operation serves

as the entry point for 2D selective scanning. Given a feature map $X \in \mathbb{R}^{H \times W \times C}$, Scan Expand generates four distinct flattened sequences by traversing the spatial dimensions from four corners (top-left, top-right, bottom-left, and bottom-right). Formally, for a pixel at (i, j) , the mapping $\phi_k(i, j)$ for $k \in 1, 2, 3, 4$ ensures that every spatial location is processed in a unique sequential order, expanding the input to a four-way sequence tensor to capture comprehensive spatial dependencies. By integrating the standard SS2D module with multi-scale convolutions, the LGF-SS2D module enables the model to dynamically balance global context and local details according to different image regions. This adaptive balance significantly improves the segmentation of complex anatomical structures.

To ensure stable gradient flow and effective feature integration, we incorporate dual residual connections:

$$F_i^2 = \text{LN}(F_i^1 + F_{\text{fused}}), \quad (4)$$

$$F_i^3 = F_i^2 \odot \sigma(\text{LN}(F_i^0)) + F_i^0. \quad (5)$$

The first residual connection (Eq. (4)) combines the preprocessed features with the fused features after layer normalization, while the second connection (Eq. (5)) introduces a gated residual path that modulates the features based on the normalized input. These enhanced residual connections facilitate multi-level feature integration and optimize gradient propagation throughout the network.

Compared to standard VSS blocks used in VM-UNet and the multi-scale design in MSVM-UNet, our LGF-VSS module offers three distinct advantages: (1) adaptive balancing of global and local features through learnable weights rather than fixed fusion, (2) multi-scale local feature extraction through parallel convolutional branches with varying receptive fields, and (3) optimized information flow through enhanced residual connections.

3.3. Multi-level Cross-scale Feature Fusion Block (MCFB)

The MCFB replaces horizontal skip connections, with its primary relational role being semantic alignment. In Mamba-based networks, encoder features (F_{enc}) have undergone selective scanning, which transforms spatial topology differently than convolutional pooling. A simple concatenation with the reconstructing decoder features (D_{dec}) assumes a spatial coherence that may not exist. The MCFB serves as a bridging mechanism. It first imports low-level details from all encoder stages and high-level semantics from the decoder. Then, it spatially and semantically aligns these features before passing them to the next decoding stage. This allows the decoder to access the entire history of the encoding process, not just features from a single corresponding level, thereby effectively mitigating the semantic gap.

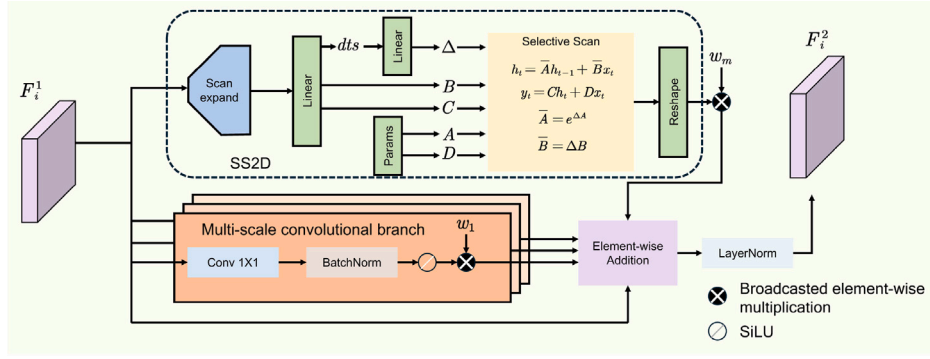


Fig. 2. Architecture of LGF-SS2D. The multi-scale convolutional branch component comprises three parallel branches, each implementing standard convolutions with kernel sizes of 1×1 , 3×3 , and 5×5 , respectively. These branches are dynamically modulated by the corresponding weight components w_1 , w_3 , and w_5 (generated by the QuadGate module) through channel-wise scaling operations. Concurrently, the state space branch is adaptively weighted by w_m to harmonize global-local feature interactions.

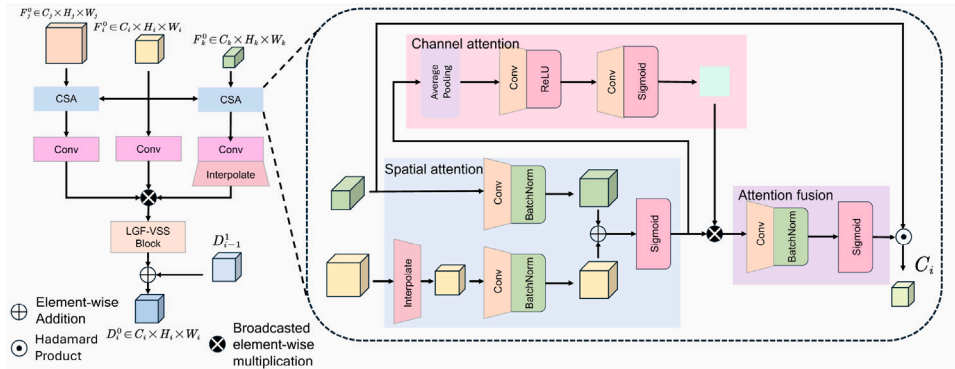


Fig. 3. Architecture of the Multi-level Cross-scale Feature Fusion Block (MCFB). The Cross-Scale Attention module (CSA) facilitates scale alignment and attention mechanisms between feature maps of different resolutions. This bidirectional feature interaction is achieved through spatial-channel dual attention mechanisms and cross-scale resampling operations, enabling comprehensive integration of semantic information from all encoder stages with the current decoder features.

To address the insufficient cross-scale feature fusion in existing Mamba-based segmentation networks, we propose the Multi-level Cross-scale Feature Fusion Block (MCFB). Unlike conventional skip connections that only provide unidirectional information flow, MCFB enables comprehensive bidirectional feature interaction through spatial-channel dual attention mechanisms and cross-scale resampling, as illustrated in Fig. 3.

For the i th decoder stage, the MCFB receives feature maps from all encoder stages $\{F_j | j = 1, 2, 3, 4\}$ and the current decoder output D_i . To clearly represent the differentiated processing for features at varying scales, we denote lower-level encoder outputs as F_j ($j < i$) and higher-level ones as F_k ($k > i$).

The first step in MCFB is to align features from different scales to the current decoder resolution. For a feature map F_j (or F_k) from encoder stage j (or k), we perform dynamic scale alignment with F_i :

$$F'_j = \begin{cases} D_{i-j}(F_j), & j < i \\ U_{k-i}(F_k), & k > i, \end{cases} \quad (6)$$

where $D_{i-j}(\cdot)$ represents cross-layer downsampling with stride 2^{i-j} using 3×3 convolutions, and $U_{k-i}(\cdot)$ denotes bilinear upsampling with scale factor 2^{k-i} . This ensures that all features are spatially aligned with the current decoder resolution F_i , facilitating subsequent attention-based feature fusion.

Unlike traditional concatenation-based fusion techniques, we employ a spatial-channel dual attention mechanism to selectively integrate cross-scale features. The spatial attention module first computes the correlation between decoder features D_i and aligned encoder features

F'_j :

$$\begin{aligned} Q &= BN(Con_{1 \times 1}(D_i)), \\ K &= BN(Con_{3 \times 3}(F'_j)), \end{aligned} \quad (7)$$

$$\Psi_{spatial} = \text{Sigmoid}(Q \oplus K),$$

where $\oplus$ denotes element-wise addition, and $\Psi_{spatial} \in [0, 1]^{H_i \times W_i}$ quantifies spatial correlations between decoder and encoder features.

The channel-wise attention component further captures semantic dependencies through a squeeze-excitation mechanism:

$$\Psi_{channel} = \text{Sigmoid}(Con_{1 \times 1}(\delta(Con_{1 \times 1}(\text{AvgPool}(\Psi_{spatial})))), \quad (8)$$

where δ represents ReLU activation, and $\Psi_{channel} \in [0, 1]^{C_j/2}$ modulates channel-wise semantic contributions.

The spatial and channel attention weights are then combined to produce a comprehensive attention map:

$$\begin{aligned} F_{att} &= \text{Sigmoid}(BN(Con_{1 \times 1}(\Psi_{spatial} \odot \Psi_{channel}))), \\ C_i &= F_{att} \odot F'_j, \end{aligned} \quad (9)$$

where $\odot$ indicates broadcasted element-wise multiplication. The refined output C_i undergoes further processing through a 3×3 convolution.

After obtaining refined features from encoder stages through the spatial-channel dual attention mechanism, we integrate them with the current decoder features. For higher-level features ($k > i$), we apply additional upsampling:

$$C'_k = U_{k-i}(Con_{3 \times 3}(C_k)). \quad (10)$$

All cross-scale features are then concatenated and processed through an LGF-VSS block before being combined with the original decoder features via a residual connection:

$$\begin{aligned} F_{concat} &= \text{Concat} \left([C_j, C_i, C'_k]_{\forall j < i, k > i} \right), \\ F_{fused} &= \text{LGFVSS} (F_{concat}), \\ D_{i+1} &= D_i + F_{fused}. \end{aligned} \quad (11)$$

3.4. Gradient statistics-based adaptive hierarchical loss

Standard Deep Supervision places auxiliary loss heads at intermediate decoder layers. These auxiliary losses are typically summed using static, pre-defined weights. However, the focus of learning at different scales shifts throughout the training process. Early training emphasizes coarse, high-level features from deeper layers, while later epochs focus on fine details from shallower layers. Static weighting cannot adapt to this shifting focus. The role of our Adaptive Loss is to monitor the learning progress at each supervised stage. If a stage becomes stuck or exhibits unstable oscillations like high gradient variance, the loss weight is automatically adjusted to stabilize its training dynamics.

As referenced in Fig. 1, we apply supervision to the outputs from each decoder stage after they have been processed through the MCFB. These outputs are first passed through auxiliary segmentation heads, each implemented as a 1×1 convolution, to generate multi-scale predictions. This hierarchical supervision scheme enables explicit training signals at different semantic levels, from low-level boundary details to high-level contextual structures. The core innovation lies in our adaptive weight allocation mechanism, which automatically adjusts the contribution of each supervision layer based on real-time gradient statistics. For the l th decoder stage at training step t , we first compute the gradient magnitude:

$$g_l^{(t)} = \frac{1}{|\theta|} \sum_{i=1}^{|\theta|} \left| \frac{\partial \mathcal{L}_l^{(t)}}{\partial \theta_i} \right|, \quad (12)$$

where $|\theta|$ denotes the total number of network parameters, and $\mathcal{L}_l^{(t)}$ is the loss for the l th decoder stage at step t . This gradient magnitude reflects the current learning status and optimization potential for each supervision layer.

To ensure stable weight updates and prevent drastic oscillations, we employ momentum-based smoothing:

$$w_l^{(t)} = \alpha \cdot w_l^{(t-1)} + (1 - \alpha) \cdot \frac{\exp(g_l^{(t)}/\tau)}{\sum_{k=1}^L \exp(g_k^{(t)}/\tau)}, \quad (13)$$

where $\alpha \in [0, 1]$ is the momentum coefficient controlling historical information retention, and $\tau > 0$ serves as a temperature parameter to control the sharpness of the weight distribution.

The final composite loss at training step t combines the adaptively weighted supervision losses with the main output loss:

$$\mathcal{L}_{total}^{(t)} = \underbrace{\sum_{l=1}^L w_l^{(t)} \cdot \mathcal{L}_l^{(t)}}_{\text{adaptive supervision}} + \lambda \cdot \underbrace{\mathcal{L}_{final}^{(t)}}_{\text{main output}}, \quad (14)$$

where $w_l^{(t)}$ represents the dynamic weight for the l th decoder stage calculated at the current iteration, and λ is a balancing hyperparameter that determines the relative importance of the main output loss. This formulation ensures that the contribution of each auxiliary head is modulated in real-time according to the optimization status of its corresponding semantic level.

To ensure the effectiveness of our adaptive loss balancing, we implement several key techniques:

(1) Gradient Normalization: We apply Z-score standardization to mitigate magnitude variations across layers.

(2) Gradient Clipping: A threshold ($g_{clip} = 1.0$) prevents exploding gradients.

(3) Hybrid Loss Function: A combination of Binary Cross-Entropy (BCE) and Dice loss addresses class imbalance.

(4) Real-time Adjustment: Gradient hooks during the backward pass enable immediate weight updates.

Unlike fixed weighting schemes used in previous works like UNet++, our adaptive approach can automatically shift focus between different decoder levels as training progresses. This dynamic balancing effectively resolves optimization conflicts between global and local features, leading to more comprehensive feature learning and improved segmentation accuracy, particularly for complex anatomical structures.

4. Experiment

4.1. Datasets

This study evaluates our proposed LGFVM-UNet on five publicly available medical imaging datasets, spanning diverse imaging modalities and anatomical structures.

Synapse Multi-organ Segmentation Dataset (Landman et al., 2015): This dataset contains 30 abdominal CT scan cases (totaling 3779 axial slices) with fine annotations for eight abdominal organs (liver, spleen, kidneys, etc.). This dataset was collected from multiple medical institutions, resulting in significant variations in scanning parameters and anatomical structures. These characteristics make it suitable for validating the robustness of multi-organ segmentation algorithms.

ACDC Cardiac MRI Dataset (Bernard et al., 2018): Provided by the 2017 MICCAI Challenge, this dataset includes 100 cardiac MRI scans with annotations for end-diastolic (ED) and end-systolic (ES) phases, covering three regions of interest: right ventricle, left ventricle, and myocardium. It encompasses healthy subjects and patients with cardiomyopathy or heart failure, ensuring pathological diversity.

ISIC Dermoscopy Image Dataset: The ISIC2017 (Codella et al., 2018) dataset comprises 2000 training images (374 malignant, 626 benign) and 600 test images, providing lesion segmentation masks and classification labels. The ISIC2018 (Codella et al., 2019) dataset expands to 2594 images (seven skin lesion categories) with annotations including lesion boundaries and histopathological classifications. Both datasets feature inter-class sample imbalance and substantial morphological variations, making them suitable for testing skin lesion analysis algorithms.

CVC-ClinicDB Colonoscopy Dataset (Bernal et al., 2015): This dataset consists of 612 polyp images extracted from 29 colonoscopy videos, offering pixel-level polyp annotations. It is characterized by significant illumination variations, specular reflection artifacts, and diverse polyp sizes and shapes, serving as a benchmark dataset for gastrointestinal lesion detection.

4.2. Implementation details

Dataset preprocessing was conducted as follows. The Synapse Multi-organ Segmentation Dataset was standardized to 224×224 pixels following VM-UNet's protocol, retaining an 18/12 training/testing split. For the ACDC Cardiac MRI Dataset, images were resampled to 256×256 resolution, and the official data partitioning was preserved. Both ISIC datasets were processed at 256×256 resolution and divided using a 60/20/20 split. The CVC-ClinicDB images were kept at their original 384×288 resolution, with a similar partitioning applied.

To ensure a fair and rigorous comparison, all baseline and state-of-the-art competitor models were re-implemented and trained from scratch within our local hardware environment under identical data preprocessing and augmentation protocols, rather than adopting results directly from their original publications.

We implemented a comprehensive data augmentation strategy tailored for medical imaging:

Table 1

Comparison of segmentation performance on the Synapse multi-organ dataset. Results show DSC (%) and HD95 metrics for each anatomical structure. Best results are highlighted in bold, while second-best results are underlined.

Methods	Average		Aorta	Gallbladder	Kidney(L)	Kidney(R)	Liver	Pancreas	Spleen	Stomach
	DSC (%)↑	HD95↓								
UNet (Ronneberger et al., 2015)	71.98	22.39	83.72	37.90	89.07	80.74	87.75	39.77	89.57	67.34
UNet++ (Zhou et al., 2018)	77.70	27.58	85.95	40.18	90.51	88.33	94.46	48.53	91.54	82.11
SwinUNet (Cao et al., 2022)	78.68	35.57	77.15	70.52	88.32	86.13	91.91	51.57	87.70	76.15
TransUNet (Chen et al., 2021)	82.42	25.54	87.72	76.52	88.87	81.38	94.99	54.49	89.18	86.25
Semi-Mamba-UNet (Ma and Wang, 2024)	83.49	26.91	87.21	70.14	89.82	86.15	95.18	62.08	92.18	85.16
VM-UNetV2 (Zhang et al., 2024)	83.56	22.29	83.01	71.68	90.17	89.30	93.49	64.81	91.81	84.57
H-vmunet (Wu et al., 2025)	86.50	<u>7.94</u>	88.63	70.93	92.34	91.24	95.91	69.19	93.81	<u>90.02</u>
MSVM-UNet (Chen et al., 2024)	<u>87.41</u>	8.91	<u>89.04</u>	70.45	<u>93.77</u>	<u>92.18</u>	96.39	<u>73.25</u>	<u>94.52</u>	89.68
LGFVM-UNet(ours)	88.74	6.65	89.13	<u>75.18</u>	94.33	93.87	<u>96.24</u>	75.00	94.55	91.61

Table 2

Quantitative evaluation on the ACDC cardiac MRI dataset showing average and per-class DSC (%) and HD95 values across right ventricle (RV), myocardium (Myo), and left ventricle (LV). Best results are highlighted in bold, while second-best results are underlined.

Methods	Average		RV	Myo	LV
	DSC (%)↑	HD95↓			
UNet (Ronneberger et al., 2015)	80.25	49.73	74.46	76.39	89.88
UNet++ (Zhou et al., 2018)	87.47	37.08	86.78	82.79	92.83
SwinUNet (Cao et al., 2022)	86.89	39.42	86.52	81.96	92.18
TransUNet (Chen et al., 2021)	88.69	35.90	86.40	85.88	93.80
Semi-Mamba-UNet (Ma and Wang, 2024)	91.96	29.16	91.03	<u>89.12</u>	95.73
VM-UNetV2 (Zhang et al., 2024)	91.03	19.87	90.46	87.67	94.95
H-vmunet (Wu et al., 2025)	<u>92.07</u>	<u>21.58</u>	<u>91.77</u>	89.09	95.34
MSVM-UNet (Chen et al., 2024)	91.76	30.60	91.13	88.80	95.35
LGFVM-UNet(ours)	92.40	21.97	92.19	90.13	<u>95.71</u>

(1) Spatial Transformations: This included rotational augmentation and elastic deformations. Rotations were applied as 90° multiples with axial flipping, or as free rotations within $\pm 15^\circ$. Elastic deformations had a 30% probability of application.

(2) Intensity Perturbations: These involved window width/level shifts ($\pm 15\%$), local brightness modulation ($\pm 20\%$), and the addition of constrained Gaussian noise ($\sigma < 0.05$).

(3) Anatomical Integrity Preservation: Third-order spline interpolation was employed to maintain the integrity of anatomical boundaries during transformations.

The training protocol remained consistent across experiments: AdamW optimizer with weight decay of $1e-4$, initial learning rate of $1e-4$ with cosine annealing, batch size of 32, and maximum 200 epochs with early stopping (patience = 15). All experiments were conducted on an NVIDIA GeForce RTX 4090 GPU using PyTorch 2.1.2. We employed our proposed gradient statistics-based adaptive hierarchical loss (as detailed in Section 3.4) with optimal hyperparameters ($\alpha = 0.8$, $\tau = 0.2$, $\lambda = 0.75$) determined through ablation studies.

4.3. Evaluation metrics

In our experiments, we adopted a comprehensive set of evaluation metrics to thoroughly assess segmentation performance. For all datasets, we used Dice Similarity Coefficient (DSC) to measure volumetric overlap between predictions and ground truth. For the Synapse and ACDC datasets, we additionally reported 95% Hausdorff Distance (HD95) to evaluate boundary delineation accuracy. For the ISIC and CVC-ClinicDB datasets, we included Mean Intersection over Union (mIoU), Accuracy (Acc), Sensitivity (Sen), and Specificity (Spe) to provide a comprehensive performance assessment.

For multi-class segmentation tasks (Synapse and ACDC datasets), we report class-wise DSC and average HD95 across all classes. For binary segmentation tasks (ISIC and CVC-ClinicDB datasets), we report all five metrics to evaluate different aspects of segmentation quality.

4.4. Comparative experimental results

Results on Synapse multi-organ dataset. As shown in Table 1, LGFVM-UNet achieves state-of-the-art performance with 88.74% DSC and 6.65 HD95. This represents significant improvements over CNN-based methods such as UNet++ (77.70% DSC, 27.58 HD95), Transformer-based approaches like TransUNet (82.42% DSC, 25.54 HD95), and recent Mamba-based methods including MSVM-UNet (87.41% DSC, 8.91 HD95). Our method shows consistent improvements across all abdominal organs. For challenging structures like the pancreas, LGFVM-UNet achieves 75.00% DSC compared to 73.25% with MSVM-UNet. Similarly, for kidneys, our method attains 94.33% and 93.87% DSC for left and right kidneys respectively, versus 93.77% and 92.18% with MSVM-UNet. This uniform enhancement across various organ scales validates our approach's balanced capability in capturing both global context and local details.

Results on ACDC cardiac dataset. Table 2 presents results on the ACDC cardiac segmentation benchmark, where LGFVM-UNet achieves 92.40% average DSC and 21.97 HD95. This represents a significant improvement over TransUNet with 88.69% DSC and 35.90 HD95. For the three cardiac structures, our method delivers superior performance with 92.19%, 90.13%, and 95.71% DSC for RV, Myo, and LV respectively. The improvement is particularly notable for the challenging myocardium region, where our method achieves 90.13% DSC compared to 85.88% with TransUNet. Although H-vmunet achieves a comparable average DSC of 92.07%, our method provides better clinical value by more accurately segmenting the thin-walled myocardium. This result confirms the effectiveness of our approach in preserving fine structural details.

Results on ISIC skin lesion datasets. On the ISIC2017 dataset (Table 3), LGFVM-UNet achieves 91.70% DSC and 84.67% mIoU, with notable improvement in sensitivity at 91.08% while maintaining high specificity at 98.30%. This balanced performance is clinically significant as it indicates reliable lesion detection without excessive false positives. Similarly, on the more challenging ISIC2018 dataset (Table 4), our method maintains superior performance with 91.35% DSC and 84.09%

Table 3

Performance comparison on ISIC2017 skin lesion segmentation dataset using five evaluation metrics (DSC, mIoU, Accuracy, Sensitivity, and Specificity). Best results are highlighted in bold, while second-best results are underlined.

Methods	DSC (%)↑	mIoU	Acc	Sen	Spe
UNet (Ronneberger et al., 2015)	77.62	63.42	90.44	87.07	91.24
UNet++ (Zhou et al., 2018)	81.60	38.55	92.79	87.07	94.08
SwinUNet (Cao et al., 2022)	83.34	71.44	94.06	80.92	97.02
TransUNet (Chen et al., 2021)	89.73	81.37	96.16	88.06	98.07
Semi-Mamba-UNet (Ma and Wang, 2024)	<u>91.58</u>	<u>84.46</u>	<u>96.87</u>	92.62	97.83
VM-UNetV2 (Zhang et al., 2024)	89.50	81.00	96.17	88.86	97.82
H-vmunet (Wu et al., 2025)	91.27	83.95	96.91	88.14	98.88
MSVM-UNet (Chen et al., 2024)	91.35	84.09	96.80	88.86	<u>98.67</u>
LGFVM-UNet(ours)	91.70	84.67	96.97	<u>91.08</u>	98.30

Table 4

Performance comparison on ISIC2018 skin lesion segmentation dataset using five evaluation metrics (DSC, mIoU, Accuracy, Sensitivity, and Specificity). Best results are highlighted in bold, while second-best results are underlined.

Methods	DSC (%)↑	mIoU	Acc	Sen	Spe
UNet (Ronneberger et al., 2015)	79.96	66.61	91.67	74.24	96.69
UNet++ (Zhou et al., 2018)	82.14	69.69	92.71	74.91	97.84
SwinUNet (Cao et al., 2022)	84.63	73.36	93.40	81.19	96.92
TransUNet (Chen et al., 2021)	88.81	79.87	95.12	86.48	97.61
Semi-Mamba-UNet (Ma and Wang, 2024)	89.96	81.75	95.61	87.82	97.86
VM-UNetV2 (Zhang et al., 2024)	89.16	80.43	95.32	86.01	98.00
H-vmunet (Wu et al., 2025)	89.76	81.42	95.59	86.31	<u>98.27</u>
MSVM-UNet (Chen et al., 2024)	<u>90.45</u>	<u>82.57</u>	<u>95.79</u>	88.98	97.76
LGFVM-UNet(ours)	91.35	84.09	96.80	<u>88.86</u>	98.67

Table 5

Comparison of segmentation performance on the CVC-ClinicDB colonoscopy dataset, evaluating boundary accuracy and region overlap with multiple metrics. Best results are highlighted in bold, while second-best results are underlined.

Methods	DSC (%)↑	mIoU	Acc	Sen	Spe
UNet (Ronneberger et al., 2015)	80.70	67.64	96.09	73.54	98.91
UNet++ (Zhou et al., 2018)	83.13	71.12	96.63	74.66	99.38
SwinUNet (Cao et al., 2022)	84.57	73.26	96.98	74.64	99.07
TransUNet (Chen et al., 2021)	89.51	81.01	97.70	88.35	98.87
Semi-Mamba-UNet (Ma and Wang, 2024)	88.59	79.53	97.54	87.26	98.81
VM-UNetV2 (Zhang et al., 2024)	89.37	80.78	97.73	87.26	99.01
H-vmunet (Wu et al., 2025)	90.18	82.11	<u>97.88</u>	<u>87.53</u>	99.18
MSVM-UNet (Chen et al., 2024)	<u>90.99</u>	<u>83.47</u>	98.11	87.18	<u>99.45</u>
LGFVM-UNet(ours)	91.03	83.55	98.11	86.58	99.55

mIoU, outperforming previous state-of-the-art methods in capturing the irregular boundaries characteristic of skin lesions.

Results on CVC-ClinicDB dataset. For polyp segmentation as shown in Table 5, LGFVM-UNet achieves 91.03% DSC and 83.55% mIoU. These results demonstrate significant improvement over Transformer-based methods such as TransUNet with 89.51% DSC, and show comparable performance to other Mamba-based methods. Despite the challenging nature of colonoscopy images with varying illumination and specular reflections, our method maintains high specificity at 99.55% while achieving competitive sensitivity. This represents an optimal clinical trade-off for polyp detection applications.

4.5. Model complexity and practical implementation analysis

The practical deployment of deep learning models in clinical environments necessitates a rigorous balance between segmentation accuracy and computational efficiency. To address the requirement for quantitative details regarding the proposed architecture, we provide a comprehensive evaluation of the total trainable parameters and the computational footprint measured in Giga Floating Point Operations per second as summarized in Table 6. These metrics were obtained using a consistent 224×224 input resolution to ensure a fair comparison across all architectural paradigms.

Standard convolutional baselines like UNet demonstrate significant computational overhead with 41.91 G FLOPs while achieving a modest Dice Similarity Coefficient of 71.98 percent. This efficiency gap is more pronounced in UNet++ which reaches 73.70 G FLOPs due to its dense skip connections, representing a calculation load over five times higher than our proposed framework while trailing in performance by 11.04 percent. Such data highlights the inherent limitations of purely convolutional stacking where increasing the receptive field leads to a disproportionate surge in temporal complexity. Conversely, Transformer-based architectures like TransUNet alleviate local bias but suffer from extreme parameter redundancy, possessing 111.42 M parameters. This scale is nearly 2.7 times that of LGFVM-UNet and poses substantial challenges for memory-constrained hardware such as portable ultrasound machines or bedside workstations. While SwinUNet offers a more efficient alternative at 5.92 G FLOPs, it sacrifices long-range dependency modeling, resulting in a lower DSC of 78.68 percent.

The emergence of State Space Models provides a more efficient path toward global context modeling. Mamba-based designs like VM-UNetV2 and Semi-Mamba-UNet demonstrate the potential for linear complexity with 2.37 G and 16.40 G FLOPs respectively. However, ultra-lightweight designs such as H-vmunet, which utilizes only 8.97 M parameters and 8.12 G FLOPs, often struggle with a capacity limitation that results in a performance gap of 2.24 percent compared to

Table 6

Quantitative comparison of model scale, computational footprint, and segmentation performance between the proposed LGFVM-UNet and state-of-the-art architectures on the Synapse dataset. Best results are highlighted in bold, while second-best results are underlined.

Methods	DSC (%) $\uparrow$	HD95 $\downarrow$	Params (M)	FLOPs (G)
UNet (Ronneberger et al., 2015)	71.98	22.39	31.04	41.91
UNet++ (Zhou et al., 2018)	77.70	27.58	26.10	73.70
SwinUNet (Cao et al., 2022)	78.68	35.57	27.17	5.92
TransUNet (Chen et al., 2021)	82.42	25.54	111.42	43.12
Semi-Mamba-UNet (Ma and Wang, 2024)	83.49	26.91	27.40	16.40
VM-UNetV2 (Zhang et al., 2024)	83.56	22.29	22.77	2.37
H-vmunet (Wu et al., 2025)	86.50	<u>7.94</u>	8.97	8.12
MSVM-UNet (Chen et al., 2024)	<u>87.41</u>	8.91	35.93	13.90
LGFVM-UNet(ours)	88.74	6.65	41.68	14.03

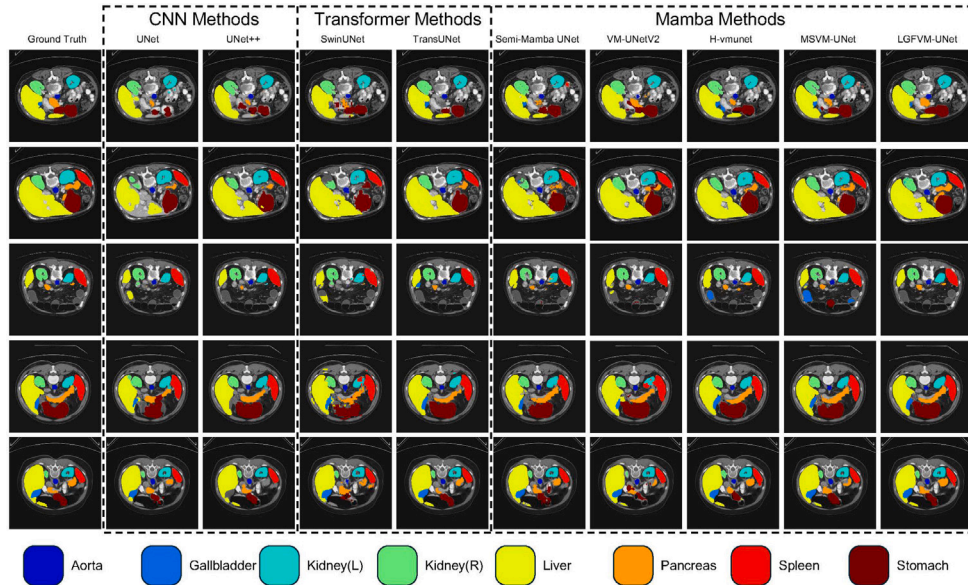


Fig. 4. Visual comparison of different methods on the Synapse multi-organ dataset. The first column represents the ground truth, while the subsequent columns show segmentation predictions from all methods mentioned in Table 1. Contours in various colors indicate the corresponding organs.

our method. In the context of critical anatomical structures, this gap translates to higher boundary errors and reduced clinical reliability.

The trade-off between performance gains and complexity is particularly evident when comparing our model to the state-of-the-art MSVM-UNet. We achieve a significant 1.33 percent DSC improvement with a marginal increase in computational footprint of less than one percent, namely 0.13 G FLOPs. While the parameter count is higher at 41.68 M, this additional capacity is essential for the QuadGate dynamic fusion mechanism and parallel multi-scale kernels which allow the model to adaptively handle diverse tissue textures. This ensures that the clinical advantage of millimeter-level boundary precision, evidenced by the reduction in Hausdorff Distance, is delivered without compromising the feasibility of real-time inference on mid-range medical workstations.

4.6. Qualitative analysis

To visually demonstrate the segmentation performance of our proposed LGFVM-UNet against existing methods, we provide qualitative comparisons using the Synapse multi-organ dataset in Fig. 4. The visualization focuses on challenging cases with complex anatomical structures and ambiguous boundaries. As shown in the figure, LGFVM-UNet produces more accurate boundary delineation compared to both Transformer-based (TransUNet, SwinUNet) and other Mamba-based (VM-UNetV2, MSVM-UNet) approaches.

Specifically, for small organs with low contrast (e.g., pancreas, shown in row 1), LGFVM-UNet captures the complete structure while

avoiding false positives in surrounding tissues. For adjacent organs with similar intensities (e.g., kidneys adjacent to liver, shown in row 2), our method successfully distinguishes between organ boundaries where other methods produce segmentation leakage.

These qualitative improvements align with our quantitative results and further validate the effectiveness of the LGF-VSS blocks in balancing global context modeling and local detail preservation, as well as the MCFB module in enhancing multi-scale feature integration. Although the numerical DSC gains may appear modest (typically 1%–2%), they correspond to clinically significant improvements in segmentation quality. For critical anatomical structures such as the pancreas, even a slight improvement in DSC often results in more continuous boundaries and a noticeable reduction in leakage into adjacent vascular tissues. In clinical applications including surgical planning and radiotherapy dose calculation, where millimeter-level accuracy is essential, such refinements carry considerable practical importance. The visualization therefore underscores the advantage of our proposed components in handling diverse anatomical structures with varying scales, shapes, and contrast levels.

Despite the overall superior performance, we conduct a transparent analysis of typical failure cases to identify the model's limitations, as illustrated in Fig. 5. In the first row, while the model accurately segments the upper aorta, it only partially identifies the smaller aorta located in the central region of the image, suggesting a sensitivity threshold for extremely small structures in specific slices. In the second row, although the LGFVM-UNet successfully delineates the Kidneys and Aorta, it almost entirely fails to segment the Pancreas. This failure

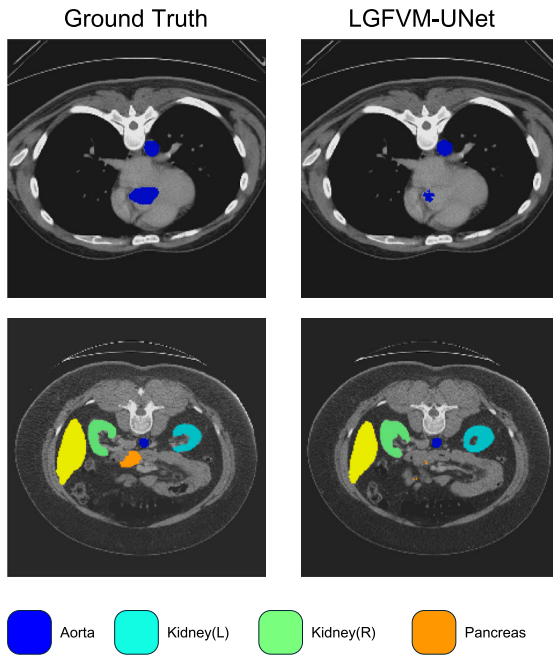


Fig. 5. Visualization of representative failure cases on the Synapse dataset. The first row demonstrates incomplete segmentation of a small central aorta. The second row shows a failure to detect the pancreas (central) despite successful segmentation of kidneys and larger vascular structures.

Table 7

Ablation study analyzing the contribution of LGF-VSS and MCFB components to segmentation performance, computational complexity, and parameter efficiency. Best results are highlighted in bold, while second-best results are underlined.

Components		DSC (%) $\uparrow$	HD95 $\downarrow$	FLOPs (G)	Params (M)
LGF-VSS	MCFB				
$\times$	$\times$	84.22	8.22	12.63	22.04
$\checkmark$	$\times$	85.52	7.48	2.56	19.12
$\times$	$\checkmark$	87.65	7.02	26.78	45.33
$\checkmark$	$\checkmark$	88.74	6.65	14.03	41.68

likely stems from the pancreas's highly irregular morphology and its extremely low intensity contrast relative to the surrounding retroperitoneal fat and gastrointestinal tissue in that particular scan. These cases indicate that while the LGF-VSS block significantly enhances local detail, extreme scale variations and indistinct semantic boundaries still pose challenges, providing a clear direction for future refinement.

4.7. Ablation studies

We conducted extensive ablation studies on the Synapse dataset to validate our design choices and analyze the contribution of each component in LGFVM-UNet.

4.7.1. Impact of key architectural components

Table 7 presents a systematic analysis of the two core components in LGFVM-UNet. The baseline model without these components achieves 84.22% DSC and 8.22 HD95. Incorporating LGF-VSS blocks alone improves performance to 85.52% DSC and 7.48 HD95, while reducing computational complexity from 12.63G to 2.56G FLOPs. This validates the effectiveness of our proposed fusion mechanism in balancing global context modeling and local feature extraction. Adding MCFB alone without LGF-VSS blocks boosts performance significantly to 87.65% DSC and 7.02 HD95, highlighting the importance of cross-scale feature fusion. However, this comes at the cost of increased computational

Table 8

Ablation study on the effect of LGF-VSS block distribution across encoder and decoder stages, comparing symmetrical and asymmetrical architectural configurations. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Encoder depth	Decoder depth	DSC (%) $\uparrow$	HD95 $\downarrow$	FLOPs (G)	Params (M)
[1, 1, 1, 1]	[2, 2, 2, 1]	<u>87.52</u>	10.98	13.87	37.68
[2, 2, 2, 2]	[1, 1, 1, 1]	<u>87.52</u>	9.26	13.95	37.73
[1, 2, 2, 2]	[2, 2, 2, 1]	87.08	9.27	13.95	41.63
[2, 2, 2, 2]	[2, 2, 2, 1]	88.74	6.65	14.03	41.68
[2, 2, 2, 2]	[2, 2, 2, 2]	87.26	9.09	14.11	41.74
[2, 2, 4, 4]	[2, 2, 2, 1]	87.19	8.66	14.12	49.2
[4, 4, 2, 2]	[2, 2, 2, 1]	86.79	10.52	14.27	42.18
[4, 3, 2, 1]	[2, 2, 2, 1]	87.32	<u>7.82</u>	14.21	38.99
[4, 4, 4, 4]	[2, 2, 2, 1]	86.85	8.24	14.36	49.7

Table 9

Ablation study on the internal branches of the LGF-VSS block. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Configuration	SSM branch	Conv branch	DSC (%) $\uparrow$	HD95 $\downarrow$
SS2D only (Standard VSS)	$\checkmark$	$\times$	84.56	8.15
CNN only (ResNet-like)	$\times$	Multi-scale	85.12	7.92
Simplified hybrid	$\checkmark$	Single 3×3	<u>86.45</u>	<u>7.34</u>
LGF-VSS (Ours)	$\checkmark$	Multi-scale	88.74	6.65

complexity to 26.78G FLOPs and parameter count to 45.33M. The full LGFVM-UNet architecture combining both components achieves the best performance with 88.74% DSC and 6.65 HD95 while maintaining reasonable computational complexity at 14.03G FLOPs and parameter count at 41.68M. This demonstrates the synergistic effect of our proposed components in enhancing segmentation accuracy.

4.7.2. Optimal distribution of LGF-VSS blocks in encoder-decoder architecture

Table 8 explores various configurations of LGF-VSS block distribution across encoder and decoder stages. Our findings reveal several important patterns. The symmetric [2,2,2,2] encoder configuration achieves optimal DSC at 88.74% compared to asymmetric allocations like [1,2,2,2] with 87.08%. This indicates that balanced feature extraction across stages is crucial for segmentation performance. Interestingly, the [4,3,2,1] configuration with progressively decreasing depth achieves better HD95 at 7.82 than uniform deep configurations, suggesting different optimal patterns for DSC versus boundary delineation metrics.

The [4,4,4,4] configuration increases parameters significantly to 49.7M but reduces DSC to 86.85%, demonstrating that excessive depth leads to feature over-smoothing. Similarly, the [2,2,4,4] structure with deeper later encoder stages delivers suboptimal performance despite increased computational cost, indicating that naively increasing depth is counterproductive.

Comparing [2,2,2,2]+[2,2,2,1] versus [2,2,2,2]+[2,2,2,2] configurations, we observe that reducing final decoder blocks from 2 to 1 improves DSC by 1.48% with minimal parameter increase of 0.06M. This suggests that the final decoder stage benefits from streamlined processing, as extensive feature refinement at highest resolution can introduce redundancy without adding discriminative information.

Based on these comprehensive ablation studies, we adopt the configuration [2,2,2,2]+[2,2,2,1] with parallel kernel sizes [1,3,5] and adaptive loss hyperparameters $\alpha = 0.8$, $\tau = 0.2$, $\lambda = 0.75$ as our final architecture, balancing segmentation accuracy and computational efficiency.

4.7.3. Detailed dissection of LGF-VSS internal structure

To rigorously evaluate the contribution of each pathway within the Local-Global Fusion Visual State Space block, we first conduct

Table 10

Ablation study on the effect of different convolutional kernel combinations in LGF-VSS blocks, evaluating the trade-off between multi-scale receptive field diversity and computational efficiency. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Kernels	DSC (%)↑	HD95↓	FLOPs (G)	Params (M)
[1, 3]	87.46	8.64	13.72	37.57
[3, 5]	88.56	7.81	13.97	37.74
[1, 3, 5]	88.74	<u>6.65</u>	14.03	41.68
[3, 5, 7]	88.49	<u>6.25</u>	14.52	42.03
[1, 3, 5, 7]	88.35	6.83	14.58	45.97

Table 11

Investigation of different feature fusion strategies within the LGF-VSS block to evaluate the impact of dynamic, input-dependent modulation. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Fusion strategy	Modulation type	DSC (%)↑	HD95↓
Fixed summation	Static	85.52	7.48
Concatenate + 1×1 Conv	Learned (Fixed)	86.88	<u>7.12</u>
QuadGate (Ours)	Dynamic (Pixel-wise)	88.74	6.65

a dissection of its internal branches as summarized in Table 9. The configuration utilizing the SS2D branch alone represents a standard Visual State Space module which relies entirely on 1D scanning for feature extraction. While this approach captures long-range dependencies effectively, the results show a decrease in Dice Similarity Coefficient to 84.56 percent and an increase in Hausdorff Distance to 8.15. This confirms that pure State Space Models often struggle with the attenuation of high-frequency local details.

Conversely, the CNN-only configuration employs parallel multi-scale convolutions without the state space branch, essentially functioning as a complex ResNet-like bottleneck. Although this setup improves local texture capture compared to the standard VSS, it lacks the global receptive field necessary for understanding organ placement, resulting in a suboptimal performance of 85.12 percent DSC. Furthermore, we evaluate a simplified hybrid model where the parallel branches are replaced by a single 3×3 convolution. While this hybrid outperforms the single-path designs, it trails the full LGF-VSS by 2.29 percent in DSC. This gap highlights the necessity of multi-scale kernels for handling the morphological diversity of anatomical structures at varying resolutions.

4.7.4. Impact of parallel convolution kernel sizes in LGF-VSS block

Table 10 analyzes different combinations of kernel sizes in the parallel convolutional branches of LGF-VSS blocks. For dual-kernel configurations, the [3,5] combination outperforms [1,3], achieving 88.56% DSC versus 87.46%. This suggests that larger kernels capture more contextual information relevant to medical image segmentation.

Among the three-kernel configurations, [1,3,5] achieves the best performance with 88.74% DSC and 6.65 HD95 with a reasonable parameter count of 41.68M. While [3,5,7] yields a slightly better HD95 at 6.25, it comes with increased computational cost and lower DSC. Expanding to four kernels with [1,3,5,7] increases parameters substantially to 45.97M without performance gains, indicating diminishing returns from additional kernel sizes. This analysis confirms that the [1,3,5] configuration represents an optimal balance between computational efficiency and segmentation accuracy.

4.7.5. Effectiveness of dynamic gating mechanism

The QuadGate module is designed to provide data-dependent arbitration between global and local feature streams. We investigate the impact of this dynamic modulation by comparing it against static fusion strategies in Table 11. In the fixed summation experiment, the weights for all branches are set to unity, which forces the network to treat global and local information with equal importance regardless of the

image content. This rigid strategy results in a DSC of 85.52 percent, as it cannot adapt to regions where either local edges or global context should dominate the representation.

A more sophisticated baseline involves concatenating the branch outputs and fusing them via a 1×1 convolution. This allows the network to learn fusion weights during the training process, yet these weights remain fixed during inference for all input pixels. Our proposed QuadGate mechanism achieves a significant improvement over this learned fixed strategy by generating pixel-wise weights that respond to the specific characteristics of each region. By assigning higher weights to convolutional branches at organ boundaries and prioritizing the Mamba branch in uniform tissue areas, the QuadGate ensures optimal feature integration, leading to the highest performance of 88.74 percent DSC and the lowest boundary error of 6.65 HD95.

4.7.6. Component analysis of MCFB

The Multi-level Cross-scale Feature Fusion Block serves as a critical bridge between the encoder and decoder hierarchies. To analyze its effectiveness, we evaluate the impact of multi-level access and the dual attention mechanism as shown in Table 12. We begin by comparing against a baseline concatenation approach that mimics the standard U-Net connection, which utilizes only single-stage encoder features and fuses them via simple concatenation. As shown in the first row of Table 12, this baseline yields a DSC of only 84.22%, indicating that without multi-level feature access and explicit attention mechanisms, the model's ability to align and integrate cross-scale features remains limited. This result underscores the necessity of the more sophisticated design adopted in our MCFB.

We first remove the multi-level inputs and feed only the corresponding encoder stage to the MCFB, mimicking the behavior of traditional skip connections. This results in a drop in DSC to 86.15 percent, demonstrating that high-level decoding stages benefit significantly from direct access to low-level features from earlier encoder stages to recover spatial precision.

Furthermore, we assess the role of the spatial-channel dual attention mechanism by replacing it with simple concatenation and 3×3 convolutions for feature alignment. Without the explicit attention-based weighting, the model achieves a DSC of 87.34 percent. The full MCFB implementation provides superior results by utilizing bidirectional resampling to ensure spatial coherence and employing dual attention to suppress noise in the encoder features while amplifying relevant semantic signals. This comprehensive alignment strategy effectively bridges the semantic gap between the linear 1D scanning of the Mamba backbone and the 2D reconstruction process of the decoder, confirming that both multi-level connectivity and adaptive attention are essential for high-fidelity medical image segmentation.

4.7.7. Comparison with advanced fusion and attention mechanisms

To further demonstrate the superiority of our proposed Multi-level Cross-scale Feature Fusion Block (MCFB) and the QuadGate dynamic fusion mechanism, we conducted a series of module-replacement experiments on the Synapse dataset. We substituted our components with several state-of-the-art (SOTA) modules from recent literature to evaluate their effectiveness in semantic alignment and local-global feature arbitration.

We compared the MCFB, which replaces standard skip connections, with two prominent fusion strategies: the Attention Gate (AG) from Attention U-Net and the Scan-Decoupled Integration (SDI) module from VM-UNetV2. As shown in Table 13, the MCFB significantly outperforms both AG and SDI. While AG effectively suppresses background noise, it lacks the ability to align features across multiple scales simultaneously. The SDI module improves multi-scale exchange, but it does not account for the semantic gap and spatial incoherence introduced by Mamba's 1D scanning. By utilizing bidirectional resampling and spatial-channel dual attention, the MCFB ensures that the encoder's local details are

Table 12

Ablation study on MCFB components, analyzing the contribution of multi-level feature access and spatial-channel attention to cross-scale alignment. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Configuration	Input level	Attention mechanism	DSC (%) $\uparrow$	HD95 $\downarrow$
Baseline concatenation	Single-stage	None	84.22	8.22
Single-level MCFB	Single-stage	Spatial-Channel	86.15	7.45
MCFB without attention	Multi-level	None	<u>87.34</u>	<u>6.98</u>
MCFB (Ours)	Multi-level	Spatial-Channel	88.74	6.65

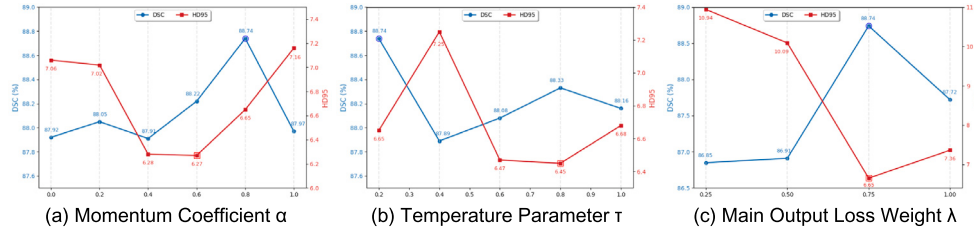


Fig. 6. Ablation studies on hyperparameters of our adaptive loss function. (a), (b), and (c) show the impact of different hyperparameters on segmentation performance measured by DSC (blue, left axis) and HD95 (red, right axis). Optimal values are marked with hollow markers. The studies demonstrate that $\alpha = 0.8$, $\tau = 0.2$, and $\lambda = 0.75$ yield the best overall performance.

Table 13

Quantitative comparison of different cross-scale fusion mechanisms on the Synapse dataset. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Fusion mechanism	DSC (%) $\uparrow$	HD95 $\downarrow$
Attention gate (Oktay et al., 2018)	85.34	7.85
SDI module (Zhang et al., 2024)	<u>87.12</u>	<u>7.15</u>
MCFB (Ours)	88.74	6.65

semantically aligned with the decoder's global context, leading to a 1.62% improvement in DSC over the SDI module.

We evaluated the QuadGate module, which dynamically balances global context (Mamba) and local details (CNN), against the widely used Convolutional Block Attention Module (CBAM) and the recent Global-Local Spatial Perception (GLSP) mechanism. The results in Table 14 indicate that a dynamic arbitration strategy is essential for medical image segmentation. While CBAM enhances features through sequential attention, it cannot adaptively mitigate the local feature attenuation inherent in pure SSMs. The GLSP module allows for parallel feature coexistence but lacks a pixel-wise weighting mechanism. In contrast, our QuadGate assigns data-dependent weights to each branch. This is particularly crucial for complex anatomical structures like the pancreas, where the model must prioritize local texture analysis over global scanning to achieve millimeter-level boundary precision.

4.7.8. Analysis of adaptive loss function hyperparameters

We conducted detailed experiments to determine optimal hyperparameters for our gradient statistics-based adaptive hierarchical loss function. Our approach implements supervision at the first three decoder stages only, with the final stage directly contributing to the main loss function. Fig. 6 and Table 15 present the results of these experiments.

Fig. 6(a) evaluates different values of the momentum coefficient α ranging from 0 to 1 while fixing τ at 0.2. The results indicate that $\alpha = 0.8$ achieves optimal performance with 88.74% DSC and 6.65 HD95, suggesting that incorporating historical gradient information with moderate retention is beneficial for stabilizing the training process. Lower values of α below 0.6 lead to suboptimal performance due to excessive fluctuations in loss weights, while $\alpha = 1.0$ effectively disables the adaptive mechanism, resulting in decreased performance.

Fig. 6(b) presents the effect of the temperature parameter τ ranging from 0.2 to 1.0 with fixed α at 0.8. The best performance is achieved

at $\tau = 0.2$ with 88.74% DSC and 6.65 HD95, indicating that a sharper weight distribution enhances the model's ability to focus on the most informative decoder levels. As τ increases, the weight distribution becomes more uniform, reducing the effectiveness of the adaptive mechanism.

Fig. 6(c) examines the impact of λ , which balances the contribution of the main output loss relative to the three auxiliary supervision signals. With $\lambda = 0.75$, the model achieves optimal performance with 88.74% DSC and 6.65 HD95. This confirms that while intermediate supervision is crucial, maintaining sufficient focus on the final prediction is necessary for optimal segmentation results. Notably, removing hierarchical supervision entirely (Table 15) significantly degrades performance to 85.82% DSC, highlighting the importance of multi-level supervision in deep networks.

5. Discussion

The LGFVM-UNet framework demonstrates robust performance in 2D segmentation, however, its architectural principles offer broader potential. For 3D medical imaging, the LGF-VSS blocks can be extended by replacing 2D scanning with 3D tri-plane scanning to capture volumetric context. In the realm of weakly supervised learning, the adaptive gating weights in our LGF-VSS could serve as attention maps to guide segmentation under sparse annotations such as scribbles or points. Furthermore, our MCFB's dual-attention mechanism is naturally suited for multi-modal fusion, for instance in CT and MRI registration, where it can align disparate feature spaces. Future work will focus on scaling this architecture to large-scale 3D datasets and exploring its efficacy in real-time surgical navigation, a scenario where Mamba's linear complexity proves most advantageous.

6. Conclusion

In this paper, we proposed LGFVM-UNet, a novel framework for medical image segmentation that effectively addresses the limitations of existing Mamba-based approaches in balancing global context modeling with local feature extraction. Our architecture incorporates three key innovations: the Local-Global Fusion Visual State Space (LGF-VSS) block, which integrates multi-directional selective scanning with parallel multi-scale convolutional branches through a QuadGate dynamic fusion mechanism; the Multi-level Cross-scale Feature Fusion Block (MCFB), which establishes comprehensive bidirectional feature

Table 14

Comparison of different fusion mechanisms for medical image segmentation. Best segmentation metrics are highlighted in bold, while second-best results are underlined.

Mechanism	Type	DSC (%) [†]	HD95 [‡]
Fixed summation	Static fusion	85.52	7.48
CBAM (Vanaja and Prakasam, 2025)	Spatial-channel attention	86.45	7.22
GLSP module (Jiang et al., 2025)	Parallel perception	87.80	6.95
QuadGate (Ours)	Dynamic Gating	88.74	6.65

Table 15

Effectiveness of the proposed gradient statistics-based adaptive hierarchical loss. Performance comparison between models with and without the proposed adaptive loss weighting mechanism demonstrates significant improvements in segmentation accuracy. Best results are highlighted in bold.

Adaptive loss	DSC (%) [†]	HD95 [‡]
Disabled	85.82	7.48
Enabled	88.74	6.65

interactions between encoder and decoder hierarchies through spatial-channel dual attention mechanisms; and a Gradient Statistics-based Adaptive Hierarchical Loss strategy that dynamically adjusts supervision weights based on real-time gradient magnitudes. Extensive experiments across five diverse medical imaging datasets (Synapse, ACDC, ISIC2017, ISIC2018, and CVC-ClinicDB) demonstrate that LGFVM-UNet consistently outperforms state-of-the-art methods, with particular advantages in segmenting challenging anatomical structures with low contrast and complex boundaries. Our comprehensive ablation studies validate the contribution of each component and identify optimal architectural configurations, confirming that the efficient long-range modeling capability of State Space Models can be effectively complemented with adaptive local-global feature fusion mechanisms to achieve superior medical image segmentation performance. Looking ahead, the principles of LGFVM-UNet hold promising potential for broader applications, including extension to 3D volumetric analysis, weakly supervised learning scenarios, and multi-modal image fusion, paving the way for future research in more complex and resource-constrained clinical environments.

CRedit authorship contribution statement

Yanbo Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization. **Zihan Mao:** Validation, Methodology, Investigation, Formal analysis. **Feiwei Qin:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Yong Peng:** Validation, Software, Data curation. **Guodao Zhang:** Validation, Methodology, Investigation. **Xugang Xi:** Visualization, Formal analysis, Data curation. **Xiaoqin Ma:** Writing – review & editing, Validation, Software. **Huanhuan Yu:** Writing – review & editing, Resources, Investigation. **Yu Zhou:** Resources, Project administration, Data curation. **Zhu Zhu:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data supporting this study are publicly available in the following repositories: Synapse dataset at <https://www.synapse.org/Synapse:syn3193805/files/>; ACDC dataset at <https://www.creatis.insa-lyon.fr/Challenge/acdc/databases.html>; ISIC2017 dataset at <https://challenge.isic-archive.com/data/#2017>; ISIC2018 dataset at <https://challenge.isic-archive.com/data/#2018>; CVC-ClinicDB dataset at <https://polyp.grand-challenge.org/CVCCLinicDB/>. The source code for this study is available at <https://github.com/NicoleDyson/LGFVM-UNet>.

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